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1. Title of the Project

"ToxiFlag: A Multi-Sample Disagreement-Based Toxicity Detection System for Hinglish Social Media Text"

2. Introduction

Social media has become one of the primary platforms for communication, such as Twitter (X), Instagram, Facebook, YouTube, and Reddit has made online communication faster accessible where millions of users share opinions, ideas, and emotions every day

However, these platforms also contain a significant amount of toxic content, including hate speech, abusive language, cyberbullying, harassment, and offensive comments. Such content negatively affects users' mental well-being and creates an unsafe online environment.

In India, many users communicate in Hinglish, a mixture of Hindi and English written in Roman script. Detecting toxicity in Hinglish is more difficult than in English because of code-mixing, spelling variations, slang, abbreviations, and informal grammar. Most existing toxicity detection systems are trained on English datasets and fail to perform well on Hinglish text.

This project proposes a Multi-Sample Disagreement-Based Toxicity Detection System that combines predictions from multiple AI models or multiple data samples. Instead of relying on a single prediction, the system identifies disagreement among predictions to detect uncertain or ambiguous cases and improve overall classification accuracy.

3. Problem Statement

Most existing toxicity detection systems are developed primarily for the English language. However, in multilingual countries like India, users frequently communicate in Hinglish, a code-mixed language that combines Hindi and English using the Roman script. Hinglish text presents several challenges for automated toxicity detection because it contains inconsistent spellings, transliterated Hindi words, slang, abbreviations, emojis, and informal grammar. For example, the same Hindi word may be written in multiple ways, making it difficult for conventional Natural Language Processing (NLP) models to understand its meaning accurately.

Although transformer-based models such as BERT and multilingual language models have improved text classification performance, they still struggle with uncertainty in code-mixed languages like Hinglish. Existing approaches rarely analyse prediction disagreement to determine whether a classification is reliable. As a result, uncertain or borderline cases are often misclassified.

Another major limitation of current toxicity detection systems is that they usually rely on a single prediction model. When the input text is ambiguous or contains mixed sentiment,

4. Objectives of the Study

The primary objective of this project is to design and develop a Multi-Sample Disagreement-Based Toxicity Detection System that can accurately identify toxic content in Hinglish social media text. The specific objectives are:

1. To collect and prepare Hinglish social media data

Collect Hinglish comments and posts from publicly available datasets.
Convert the text into a clean and consistent format for better analysis.
Prepare the dataset for training and testing the model.

2. To develop a toxicity detection model

Build an AI model that can identify whether a social media comment is toxic or non-toxic.
Train the model using Natural Language Processing (NLP) and transformer-based techniques.

3. To improve the accuracy of toxicity detection

Reduce incorrect classifications of toxic and non-toxic comments.
Increase the overall reliability and consistency of the detection system.

4. To evaluate the performance of the proposed system

Compare the performance of the proposed disagreement-based system with a traditional single-model approach.
Analyse the results to determine whether the proposed method provides better performance.

5. To support safer online communication

Help social media platforms detect harmful content more effectively.
Support moderators in identifying abusive, offensive, and hateful comments.
Contribute to creating a safer and more positive online environment for users.

5.Scope of the Study

The scope of this study is to develop an intelligent system that can detect toxic content in **Hinglish social media text** using Artificial Intelligence (AI) and Natural Language Processing (NLP). The project mainly focuses on identifying abusive, offensive, hateful, or harmful comments written in a mixture of Hindi and English using Roman script.

1. It classifies social media comments or posts into **toxic** and **non-toxic** categories, helping to identify harmful online content.
2. The project uses **Natural Language Processing (NLP)** and **transformer-based deep learning models** to understand the meaning and context of code-mixed language.
3. A **multi-sample disagreement mechanism** is used to compare multiple predictions and identify uncertain or ambiguous cases, improving the reliability of toxicity detection.
4. The system is designed to reduce incorrect classifications, including **false positives** (safe comments marked as toxic) and **false negatives** (toxic comments marked as safe).
5. The model is trained and tested using publicly available Hinglish toxicity datasets and evaluated with standard performance metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.

6. Review of Literature

Recent advances in Natural Language Processing (NLP) have significantly improved toxicity detection using machine learning and deep learning techniques. Traditional classifiers such as Naïve Bayes, Logistic Regression, Random Forest, and Support Vector Machines have been widely used for text classification.

Transformer-based models such as BERT, RoBERTa, XLM-RoBERTa, IndicBERT, **and** MuRIL have achieved higher accuracy by learning contextual word representations. Researchers have also explored multilingual models for detecting hate speech and offensive language.

However, studies indicate that code-mixed languages like Hinglish remain challenging due to inconsistent spellings, transliterated Hindi words, and limited annotated datasets. Recent research suggests that ensemble learning and uncertainty estimation can improve prediction reliability, but disagreement-based toxicity detection for Hinglish remains relatively unexplored.

7. Research Gaps

Although many studies have been conducted on toxicity detection using Artificial Intelligence (AI) and Natural Language Processing (NLP), several important challenges still exist, especially for Hinglish (Hindi-English code-mixed) social media text. The following research gaps motivate the need for this study.

1. Limited Support for Hinglish Language

Most existing toxicity detection models are trained on English-only datasets.

They do not perform well on Hinglish because it mixes Hindi and English in the same sentence.

The lack of Hinglish-focused models reduces detection accuracy.

2. Difficulty in Understanding Code-Mixed Text

Hinglish sentences often contain a mixture of Hindi and English words written in Roman script.

The same Hindi word can be written in different spellings (for example, *achha*, *acha*, *accha*), making it difficult for AI models to understand the correct meaning.

Existing systems struggle to process these variations effectively

Limited Real-Time Moderation Support

Many research models are evaluated only in offline environments.

They are not optimized for real-time content moderation on fast-moving social media platforms.

8. Need for More Reliable and Explainable AI

Most existing models only provide the final classification (toxic or non-toxic) without explaining why the decision was made.

Lack of transparency reduces user trust and makes it difficult for moderators to review decisions.

8. Proposed Methodology

The proposed methodology describes the step-by-step process used to develop the Multi-Sample Disagreement-Based Toxicity Detection System for Hinglish Social Media Text. The system uses Artificial Intelligence (AI), Natural Language Processing (NLP), and transformer-based deep learning models to detect toxic comments more accurately. Instead of relying on a single prediction, the system compares multiple predictions and uses their level of agreement to make a more reliable final decision.

Step 1: Data Collection

Collect publicly available Hinglish social media datasets from sources such as HASOC, Kaggle, and OLID.

The dataset will contain comments labeled as toxic or non-toxic.

Ensure that the dataset includes different forms of abusive language, hate speech, offensive language, and normal comments.

Step 2: Data Preprocessing

The collected data will be cleaned and prepared before training the model.

Removing URLs, hashtags, mentions, and unnecessary symbols.

Removing duplicate and missing records.

Converting all text to lowercase.

Handling emojis and punctuation where appropriate.

Tokenizing the text into words or subwords.

Normalizing different spellings of Hinglish words (for example, *acha*, *accha*, and *achha*).

Step 3: Text Representation

Convert the cleaned text into numerical representations using transformer-based language models.

Models such as IndicBERT, MuRIL, or XLM-RoBERTa will be used to generate contextual embeddings.

These embeddings help the system understand the meaning and context of Hinglish sentences.

Step 5: Multi-Sample Disagreement Analysis

This is the main contribution of the proposed system.

- Generate multiple predictions using different models or different augmented versions of the same input.

- Compare the predictions from all models.

- Measure the level of agreement or disagreement among the predictions.

- If all models agree, the prediction is considered highly reliable.

- If the models disagree significantly, the comment is marked as uncertain and additional decision rules are applied before assigning the final label.

Step 6: Toxicity Classification

Based on the disagreement analysis, classify each comment as Toxic **or** Non-Toxic.

The final decision is made using the combined output of all predictions rather than relying on a single model.

This helps improve overall detection performance.

9.Tools, Techniques, Software, or Instruments to be Used

Programming Language

Python

Development Environment

Jupyter Notebook
Visual Studio Code

Libraries

Pandas
NumPy
Scikit-learn
TensorFlow
PyTorch
Matplotlib
spaCy

Dataset Sources

Kaggle
HASOC Dataset
OLID Dataset
Hinglish Offensive Language Dataset

10. Data Collection and Analysis Methods

Data Collection

The data for this project will be collected from publicly available Hinglish social media datasets, such as HASOC and Kaggle. The dataset will contain comments labeled as toxic or non-toxic. Before using the data, it will be cleaned by removing duplicate records, links, special characters, and unnecessary symbols. The text will then be prepared so that the AI model can understand it easily.

Data Analysis

After preprocessing, the data will be analyzed using Natural Language Processing (NLP) and an AI model. The model will predict whether a comment is toxic or non-toxic. Multiple predictions will be compared using a multi-sample disagreement approach to improve the accuracy of the results. Finally, the system's performance will be measured using Accuracy, Precision,

Recall, F1-score, and Confusion Matrix.

11. Expected Outcomes / Deliverables

The expected outcome of this project is to develop an intelligent system that can accurately detect toxic content in Hinglish social media text. The proposed system will help improve online safety by identifying harmful comments more reliably.

Expected Outcomes

Develop an AI-based system that can classify Hinglish social media comments as toxic or non-toxic.

Improve the accuracy of toxicity detection by using a multi-sample disagreement approach.

Reduce incorrect predictions, such as false positives and false negatives.

Detect abusive, offensive, and hate speech more effectively in Hinglish text.

Prepare complete project documentation, including the source code, trained model, final report, and presentation.

Deliverables

At the end of the project, the following deliverables will be available:

A trained toxicity detection model.

A working prototype of the proposed system.

Clean and preprocessed Hinglish dataset.

Project report and documentation.

Final project presentation (PPT).

12. Work Plan / Time Schedule

Phase	Activity	Duration
Phase 1	Literature Survey	1 Week
Phase 2	Requirement Analysis and Dataset Collection	1 Week
Phase 3	Data Cleaning and Preprocessing	2 Week
Phase 4	Feature Engineering and Text Representation	1 Week
Phase 5	Model Development	2Week
Phase 6	Multi-Sample Disagreement Implementation	1Week
Phase 7	Testing and Performance Evaluation Documentation and Final Report Preparation	2 Week

13. References / Bibliography (IEEE referencing style)

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